

MechWork.Rmd

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Introduction

This file investigates questions about world aggregate mechanical work. The questions were raised by Tiago Domingos regarding Figure 7 of the paper P. E. Brockway, M. K. Heun, Z. Marshall, E. Aramendia, P. Steenwyk, T. Relph, M. Widjanarko, J. J. Kim, A. Sainju, and J. Irtube. A country-level primary-final-useful (CL-PFU) energy and exergy database: Overview of its construction and 1971–2020 world-level efficiency results. Environmental Research: Energy, 1(025005):1–29, jun 2024.

Data

Gather and save data from the database.

```
conn <- PFUPipelineTools::get_mexerdb_conn(user = "dbcreator")
world_mats <- PFUPipelineTools::pl_filter_collect(
  db_table_name = "PSUTReAllChopAllDsAllGrAll",
  Country == "World",
  conn = conn,
  collect = TRUE)
world_mats |>
  saveRDS(file = "world_mats.rds")
```

Analysis

Load data.

```
getwd()
```

```
## [1] "/Users/mkh2/github/PFUPipeline2/report_scripts/Brockway Paul"
```

```
world_mats <- readRDS(file = "world_mats.rds")
```

```
world_mats |>
  dplyr::filter(Dataset == "CL-PFU IEA+MW",
    EnergyType == "E",
    LastStage == "Useful",
    IncludesNEU == FALSE) |>
  dplyr::select(-R, -U, -U_EIOU, -U_feed, -r_EIOU, -V, -S_units) |>
  dplyr::arrange(Year, ProductAggregation)
```

```
## # A tibble: 305 x 14
```

##	Dataset	ValidFromVersion	ValidToVersion	Country	Method	EnergyType	LastStage
##	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
##	1 CL-PFU I~	v2.0	v2.0	World	PCM	E	Useful
##	2 CL-PFU I~	v2.0	v2.0	World	PCM	E	Useful
##	3 CL-PFU I~	v2.0	v2.0	World	PCM	E	Useful

```

## 4 CL-PFU I~ v2.0          v2.0          World  PCM  E          Useful
## 5 CL-PFU I~ v2.0          v2.0          World  PCM  E          Useful
## 6 CL-PFU I~ v2.0          v2.0          World  PCM  E          Useful
## 7 CL-PFU I~ v2.0          v2.0          World  PCM  E          Useful
## 8 CL-PFU I~ v2.0          v2.0          World  PCM  E          Useful
## 9 CL-PFU I~ v2.0          v2.0          World  PCM  E          Useful
## 10 CL-PFU I~ v2.0         v2.0          World  PCM  E          Useful
## # i 295 more rows
## # i 7 more variables: IncludesNEU <lgl>, Year <int>, ChoppedMat <chr>,
## #   ChoppedVar <chr>, ProductAggregation <chr>, IndustryAggregation <chr>,
## #   Y <list>

```